InkProf | v1.0.1 | 2026-10-11

Your image deserves the right colours

You chose the light, shaped the image and worked on every nuance. When the image reaches paper, that work should come with it.

Carry the image's expression all the way to print

The warmth of skin tones, the nuances of a landscape and the balance of a black-and-white image are part of what you want to convey. A considered colour workflow gives you a stronger basis for preserving that intended expression in print. For the photographer, it brings confidence to the final, decisive step from image file to finished photograph. [1]

Make creative decisions with greater confidence

Knowing how your printer, paper and ink reproduce colour makes it easier to choose a paper and assess the image before printing. A suitable profile can also be used for soft proofing: simulating the print on a profiled display in your image editor. This helps you make informed adjustments before committing paper and ink. [1, 2]

Make room for the image and the client

Every additional test print takes time: adjust, print, wait and assess again. A controlled and verified workflow can reduce unnecessary trials and the need to make new prints. The practical benefit is more time for your images and clients, with less material spent searching for the cause of a colour problem.

Build confidence through results people can understand

For a printing business, colour accuracy is also about cooperation. A clear reference, recorded settings and measured results make it easier to explain what has been delivered and what a client can expect. They provide a shared basis for assessing quality. [3]

InkProf connects colour targets, spectrometer measurements, profiling and verification. You get both an ICC profile for your printing setup and documented evidence for assessing the result.

EVERYDAY BENEFITS

More control. Less guesswork.

Make repeat work easier to follow

When a client orders more prints or adds to a series, you want to know which conditions produced the earlier result. InkProf collects materials, settings, measurements and iterations in one project. This helps you reproduce the working process and identify changes that need investigation.

Open source brings confidence and transparency

Open source gives you the confidence of being able to follow and inspect how InkProf works. The workflow is visible: from colour patches and measurement to calculation and documented results. InkProf is not a closed “black box”. This transparency helps you understand how difficult colours are handled, right up to the limits of your chosen printer, paper and ink. The code and saved evidence let you investigate differences and make informed decisions.

Choose quality you can describe

Measurement gives photographers and printing businesses a concrete language for discussing quality. InkProf reports differences between the desired colour, the profile's prediction and the measured print. Documented standards references provide context; the measurement certificate records the scope, results and decision.

A connected colour workflow

InkProf handles printer profiling and its documentation. Combine it with a calibrated and profiled display for image assessment, correct print colour management and suitable viewing light. The profile describes a particular combination of printer, paper, ink and settings. [1, 2]

Sources behind the motivation

Web sources checked on 2026-10-03. Adobe describes predictable colour output, soft proofing and the link between profiles and printing conditions. Fogra describes predictable workflows and objective colour assessment. The International Color Consortium (ICC) promotes open, vendor-neutral colour management. [1-4] Applying these ideas to InkProf and explaining the potential to reduce trials is our synthesis, not a measured saving or an endorsement of InkProf by these organizations.

[1] Adobe - Understanding color management

[2] Adobe - Print with color management in Photoshop

[3] Fogra - Researching Prepress

[4] International Color Consortium - mission

THE WORKFLOW

From colour target to decision

InkProf is a project-based MATLAB app for profiling RGB-driven printers. It guides the work and preserves the connection between inputs and results.

01 Create or open a project

Record the printer, paper, ink, print settings and user in one place. Create or import the RGB target and choose a layout that suits the paper and measurement equipment.

02 Save TIFF16 and print separately

The app saves 16-bit TIFF files and matching measurement inputs. Print in your usual application and allow the sheets to dry. You can close the project and resume later.

03 Measure and review

Measure with a high-quality spectrometer, such as the i1Pro 2. The app handles instrument measurement and saved measurement revisions. Review and accept the correct revision before locking the inputs.

04 Build and check the ICC profile

ArgyllCMS builds the profile. The profiling process considers both colour accuracy and smooth tonal transitions, helping gradients reproduce evenly in print. Numerical checks examine model fit and colour transformations. A generated profile still needs assessment with a real print.

05 Verify a real print

Save the verification target, print it and measure it. The analysis separates the difference from the desired colour from the difference against the model's prediction. This helps you decide what needs closer investigation.

06 Approve, document or improve

Review the measured result and approve the profile for its stated use, or add colours for another iteration. When improvement stops, verify the exact final profile with a fresh control print before measured delivery. Profiles can also be compared at shared sample points. Delivery without a new verification print is explicitly labelled numerical-only.

TIFF16 is saved for separate printing. Measurement takes place in the app. Save the ICC and PDF/HTML certificate where you need them, while the project preserves the evidence and history.

FEATURES AND DOCUMENTED DELIVERY

Work you can trace

Projects that remember

Each project has its own folder. JSON records workflow state, settings, iterations and result relationships. The app blocks dependent operations when inputs are missing or have changed. You can open the results log and move the complete project folder between computers.

Improvement with a clear purpose

Select additional patches from verification errors, image colours or a clicked gamut area, then print and measure to refine. Paper suggestions adapt to patch count and equipment. Compare iterations in a 2D slice at L*=50 or in 3D.

A measurement certificate to hand over

PDF and HTML document measurements, the current ICC and reviewer decisions. ColorChecker verification adds named reference and measured swatches with Delta E00. ICC, certificate, HTML, evidence and gradient JPG previews are saved in one movable folder. Soft proof is a model prediction, distinct from measured print accuracy.

Standards references supporting assessment

ISO stands for International Organization for Standardization. InkProf documents comparison references including ISO 12647-7:2016 through MediaStandard Print 2018, published by Bundesverband Druck und Medien (bvdm), the German printing and media industry association. Selected limits provide context for assessment; InkProf's custom RGB targets do not by themselves constitute a complete standards test or ISO certification.

Open source and technical foundation

InkProf is open source under GNU GPL version 3 or later. It uses base MATLAB without mandatory add-on toolboxes, together with Python and ArgyllCMS. External dependencies have their own licences. The code, licence and third-party references are available on GitHub.

Test environment and use

The software has been tested on macOS. The complete InkProf workflow has not been verified on Windows or Linux. The program is provided without warranties under the GPL; users are responsible for assessing profiles and prints for their intended use. See the licence and the certificate's legal appendix for the terms.

MediaStandard Print 2018, Table 30, page 50

InkProf – source and release information

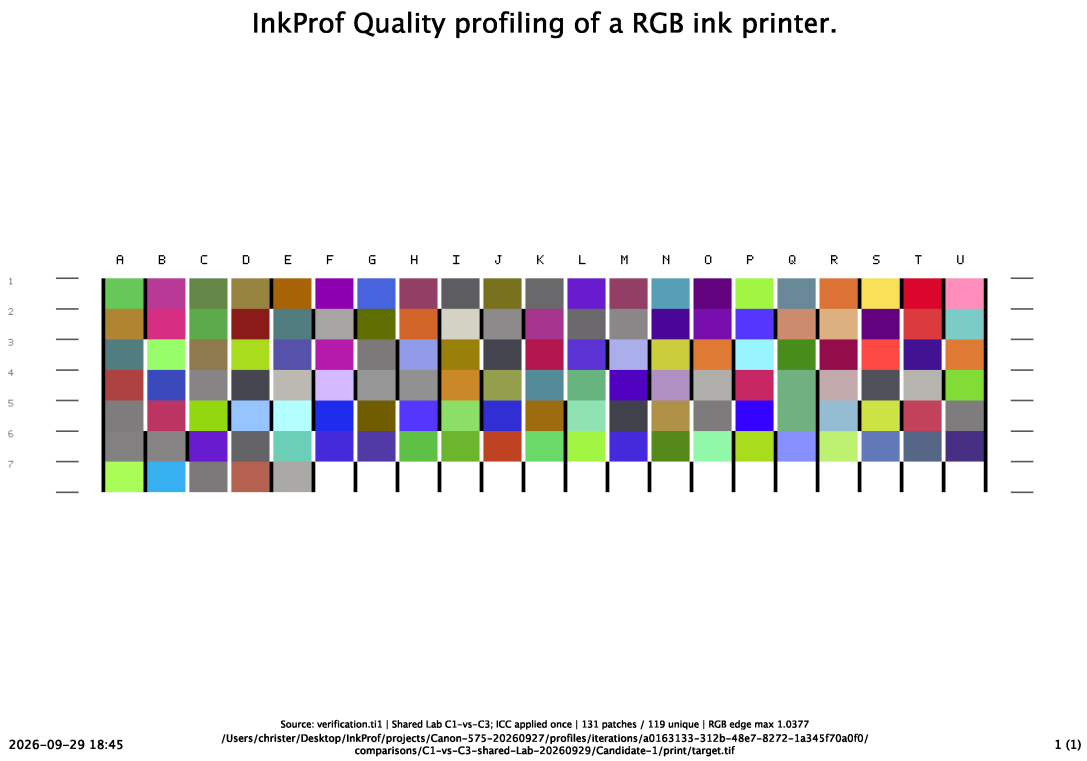
Licensing and third-party references

christer@borgasundsfotografiska.se

VISUAL EXAMPLES FROM INKPROF

From colour patches to gamut

A printed target provides measurements. The following page shows a cross-section of the colour gamut described by the resulting profile.



Patch target. An example with 131 patches, including repetitions, contrast markers and position labels. Do not use this reduced image as a measurement print.

Measurement and model complement each other

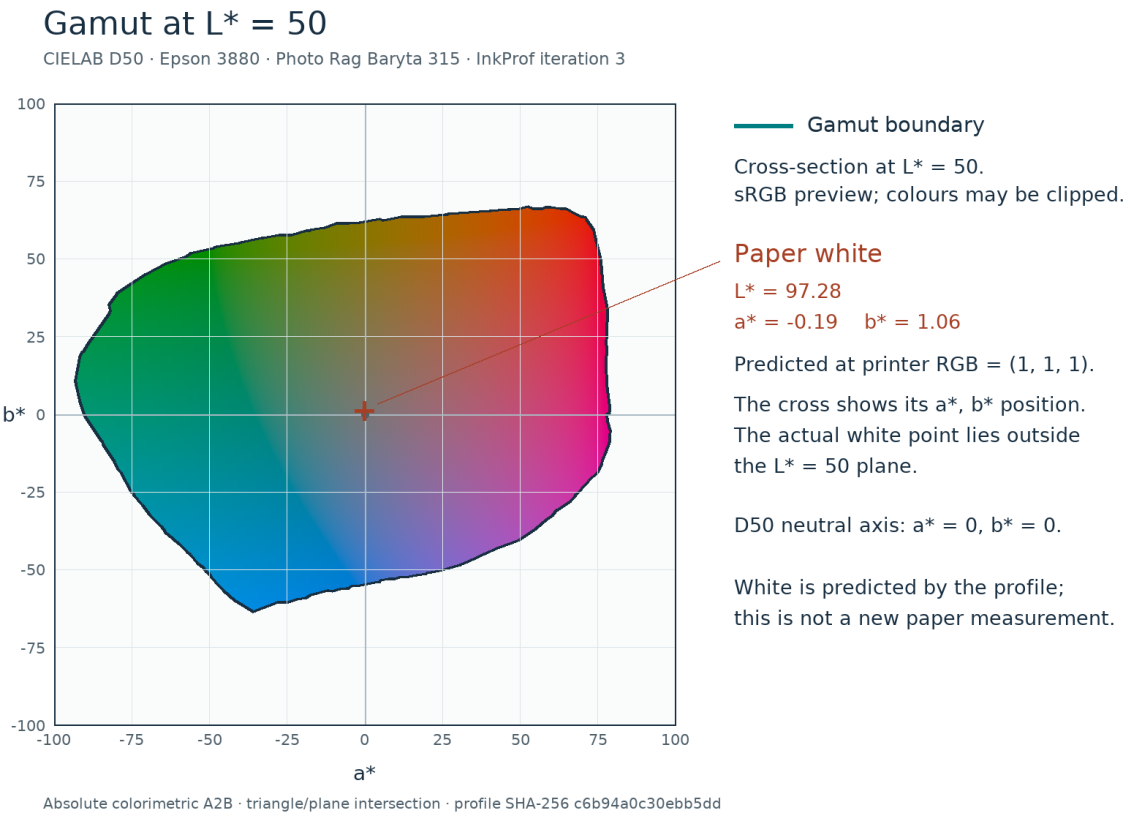
A model makes colour conversion possible. Measurement shows what actually reached the paper. Together, they support the next decision: use the profile, check a deviating patch or add more colours in another iteration.

Existing InkProf target example, 2026-09-29. The illustration is not a new validation result.

ICC GAMUT | CIELAB D50 | L* = 50

The colour gamut at L* = 50

A CIELAB cross-section shows the profile gamut at a fixed lightness. The a* axis is horizontal and b* is vertical.



Epson Stylus Pro 3880, Photo Rag Baryta 315, InkProf iteration 3. ICC-derived absolute colorimetric gamut; sRGB colour previews.

Read the gamut boundary

The contour is the intersection of the ICC-derived gamut surface with the L* = 50 plane. It shows the described colour extent at this lightness, not the entire gamut or a new measurement.

Paper white is shown separately

The cross marks the a*/b* projection of the profile-predicted paper white: L* = 97.28, a* = −0.19, b* = 1.06. Its actual lightness lies outside the L* = 50 slice. The white point is calculated at printer RGB = (1, 1, 1).

ArgyllCMS A2B, absolute colorimetric, Lab D50. Triangle/plane intersection; no convex hull. Profile SHA-256: c6b94a0c30ebb5dd… Screen colours may be clipped to sRGB.

ArgyllCMS: ICC gamut surface

MEASUREMENT QUALITY AND EARLY CHECKS

Catch errors before they carry forward

InkProf helps handle measurement errors and check whether readings are plausible before they are used to build the profile.

Remeasure a faulty reading

Reading errors can be followed by another scan of the row. The app tracks confirmed readings and shows which rows remain before the measurement can be saved.

Review plausibility and deviations

Measured values, repeated colour patches and scan direction help reveal deviations. Results are reviewed before acceptance so suspicious readings can be investigated or measured again.

Check the print before a long measurement

New targets have separate red, green and blue control squares. These are measured first and compared with an approved reference for the same printing conditions. Large deviations stop further measurement and can reveal incorrect colour management, even when an Adobe Photoshop preset was used.

Keep decisions traceable

A completed RGB check with large deviations can be overridden with a written reason, for example when a paper needs investigation. Readings, thresholds and the decision are retained. Older targets without control squares can still be measured.

These checks reduce the risk of carrying faulty prints or readings forward. A passed check still needs to be followed by profile review and verification.

v1.0.1 | NEW OPTIONS

Understand your print system

The limits of the print system

Colour accuracy is limited by what the printer, paper, ink and print settings can physically reproduce. InkProf helps assess the achievable result and documents remaining colour differences. A lack of further improvement does not by itself prove that a physical limit has been reached; the selected profile still needs verification with a measured print.

One project definition

Printer, paper and finish, dye or pigment ink, printer coating and drying time are recorded centrally and included in the measurement certificate.

Correction for optical brighteners

Correction for optical brighteners in paper can be selected when needed. The selected correction is documented with the profile and results.

More dark colour samples for matte paper

An editable shadow strategy proposes up to 48 extra dark fitting patches per iteration and denser numerical shadow tables. Colour selection follows the current profile’s black-to-white range.

Paper according to need

Cut-sheet or roll suggestions adapt to patch count and your measurement sled dimensions. Suggestions and limits remain editable.

Support throughout the work

The user handbook supports the complete profiling process, from planning and printing to assessment and final verification. Practical guidance helps you decide when to refine, when to stop and how to evaluate the finished result.

Keep using existing instruments

X-Rite has ended updates and support for i1Pro 2 and earlier generations. InkProf uses ArgyllCMS independently of i1Profiler. Compatible MATLAB, Python and ArgyllCMS versions run natively on Apple Silicon or Intel. On Apple Silicon, no Rosetta is needed, supporting continued use after its removal. The i1Pro 2 workflow is tested on current macOS; future macOS versions require verification.

Stable patch release

Source release 1.0.1, published 11 October 2026. Saved targets and measurements are preserved. Assess improvement through printing and measurement.

Copyright 2026 Christer Törnkvist. InkProf: GPL-3.0-or-later. External components retain their own licences. Source, notices and validation status are on GitHub.

X-Rite: i1Pro 2 EOL

X-Rite: i1Pro 1 support

christer@borgasundsfotografiska.se